The empirical recurrence for OEIS A302083, recovered from its tail

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Abstract

OEIS A302083 records "Empirical recurrence of order 92" and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 128$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 2$, so the recurrence is proved for every $n \geq 94$. Its last coefficient is $c_{92} = 1 \neq 0$, so no further tail satisfies anything shorter; any order-92 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry's recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A302083 is "Number of 4 X n 0..1 arrays with every element equal to 0, 1, 3 or 4 horizontally or antidiagonally adjacent elements, with upper left element zero.". It has offset 1 and begins

$$8, 34, 115, 265, 692, 2838, 10526, 36080, \ \ldots$$

The entry states:

> Empirical recurrence of order 92 (see link above).

As of the "Last modified" line on the live entry (Oct 26 2025 03:52:06, revision 10) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry's condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u, and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) \ = \ \iota^{\top} M^{\,j(n)} \tau$$

for a linear reindexing j fixed by the entry's shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 128$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most* $S' = 128$.

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \ldots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S'+1$ consecutive values of a. $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 92: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 2$ is the first whose minimal order is exactly the stated 92.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most* S' *is determined, together with its minimal recurrence, by any* $2S'$ *consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 256$ exact terms from $n = 2$ on, it returns order 92 — the order the entry states — with integer coefficients:

$$a(n) \;=\; \sum_{i=1}^{92} c_i \, a(n-i),$$

c_1	5	c_{24}	−17887	c_{47}	−655656	c_{70}	−12848
c_2	−4	c_{25}	−29365	c_{48}	−131935	c_{71}	26036
c_3	−1	c_{26}	37520	c_{49}	−1373589	c_{72}	4703
c_4	−7	c_{27}	29273	c_{50}	−179460	c_{73}	−12763
c_5	−8	c_{28}	−32675	c_{51}	−1509055	c_{74}	−3528
c_6	−20	c_{29}	−44622	c_{52}	54187	c_{75}	7559
c_7	40	c_{30}	29056	c_{53}	−930284	c_{76}	575
c_8	14	c_{31}	−27474	c_{54}	69644	c_{77}	−3742
c_9	−110	c_{32}	−126512	c_{55}	−407348	c_{78}	−440
c_{10}	−12	c_{33}	−64634	c_{56}	229286	c_{79}	2110
c_{11}	886	c_{34}	−62268	c_{57}	−60002	c_{80}	−100
c_{12}	100	c_{35}	60002	c_{58}	62268	c_{81}	−886
c_{13}	−2110	c_{36}	−229286	c_{59}	64634	c_{82}	12
c_{14}	440	c_{37}	407348	c_{60}	126512	c_{83}	110
c_{15}	3742	c_{38}	−69644	c_{61}	27474	c_{84}	−14
c_{16}	−575	c_{39}	930284	c_{62}	−29056	c_{85}	−40
c_{17}	−7559	c_{40}	−54187	c_{63}	44622	c_{86}	20
c_{18}	3528	c_{41}	1509055	c_{64}	32675	c_{87}	8
c_{19}	12763	c_{42}	179460	c_{65}	−29273	c_{88}	7
c_{20}	−4703	c_{43}	1373589	c_{66}	−37520	c_{89}	1
c_{21}	−26036	c_{44}	131935	c_{67}	29365	c_{90}	4
c_{22}	12848	c_{45}	655656	c_{68}	17887	c_{91}	−5
c_{23}	14516	c_{46}	0	c_{69}	−14516	c_{92}	1

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every* $n \geq 94$*, and it is the recurrence OEIS A302083 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 2$, whose minimal recurrence it is.

For the identification, write q_{min} for the characteristic polynomial of that minimal recurrence, $x^{92} - c_1 x^{92-1} - \cdots - c_{92}$. Its constant term is $-c_{92} = -1$, which is not zero, so $q_{\text{min}}(0) \neq 0$ and q_{min} has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since q_{min} has none, no component of a dies out, and the minimal recurrence of every further tail is again q_{min} — the order cannot drop below 92 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 92, so $\deg q = 92$, and it is monic. It holds from some index t on. From $\max(t, 2)$ on, the sequence satisfies both q and q_{min}, and the minimal polynomial of that tail is q_{min} by the previous paragraph; therefore $q_{\text{min}} \mid q$. Both are monic of degree 92, so $q = q_{\text{min}}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 26 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry's English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 92, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry's.

Third, the recovered recurrence was evaluated directly on the entry's own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is 1.

Remark 4. The conjecture's own statement was never read. The entry pins it down to the family of order-92 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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